Permutations and Combinations

Section 1: Fundamental Counting and Linear Permutations

1 In how many ways can 5 different books be arranged on a shelf?

a. 120

b. 24

c. 60

d. 720

2 Evaluate the number of permutations of 7 distinct items taken 3 at a time, ${}^{7}P_{3}$.

a. 210

b. 35

c. 42

d. 840

3 How many 4-digit even numbers can be formed using digits $\{1, 2, 3, 5, 7, 8\}$ without repetition?

a. 120

b. 60

c. 240

d. 180

Section 2: Permutations with Repetitions and Identical Items

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4 Find the number of distinct permutations of all the letters in the word PARALLEL.

a. 3360

b. 6720

c. 1680

d. 40320

5 How many 6-letter arrangements of PEPPER begin with the letter P?

a. 30

b. 60

c. 20

d. 120

6 How many 5-digit codes can be formed from digits $\{3, 3, 3, 7, 7\}$?

a. 10

b. 20

c. 5

d. 120

Section 3: Combinations and Committee Selections

7 A committee of 4 is to be chosen from 9 people. How many distinct committees can be formed?

a. 126

b. 3024

c. 84

d. 252

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8 A team of 5 is selected from 6 boys and 4 girls. How many teams contain exactly 3 boys and 2 girls?

a. 120

b. 60

c. 210

d. 100

9 From 10 points on a plane where no three are collinear, how many distinct triangles can be drawn?

a. 120

b. 720

c. 90

d. 45

Section 4: Restricted Arrangements and Separations

10 In how many ways can 4 boys and 3 girls sit in a row if all 3 girls must sit together?

a. 720

b. 5040

c. 144

d. 288

11 In how many ways can 5 people sit around a circular dinner table?

a. 24

b. 120

c. 60

d. 12

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12 A committee of 4 is selected from 5 men and 5 women. How many committees contain at least one woman?

a. 205

b. 210

c. 195

d. 150

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